Laya v2: Problem and Solution Brief

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Context

Laya v2 is a proposed redesign of our project for Velocity's Blockchain Innovation Challenge powered by Stellar (Nov 2–18, 2026). It replaces v1.1, which paid off existing predatory loans at a legal value. That model needed lenders to accept less than they were owed, so we dropped it.

- **Problem space:** 1, Access to Fair Financial Assets and Affordable Financing; target area 5, "loans that fit real income."
- **Team:** four first-year students, building on Stellar testnet with AI coding tools (allowed by the organizers).
- **Status:** a proposal for the team to agree on. Nothing in code or the plan doc has changed yet.
- **Judging:** judges think like investors and want a live demo plus a pitch. Showcase Nov 16, finals Nov 18.

The problem

Filipino domestic workers borrow at predatory rates to pay pre-departure costs, and much of what they borrow for is money they never legally owed.

Who: a Filipina domestic worker (Maria, fictional) preparing to work in Hong Kong. Before she leaves she is billed for training, a medical exam, documents, a visa and a "placement fee." Her first salary is months away.

What the law says:

- The Hong Kong employer is responsible for the helper's document costs for taking up the job and for her pre-entry medical exam. If she pays, she can claim it back (HK Labour Department, HK Government).
- The Philippines has a zero placement fee policy for domestic workers, though agencies may charge training and medical fees (SCMP).
- A Hong Kong agency may charge at most 10% of the first month's wages, and only after placement (Employment Agencies Administration).

What happens in practice:

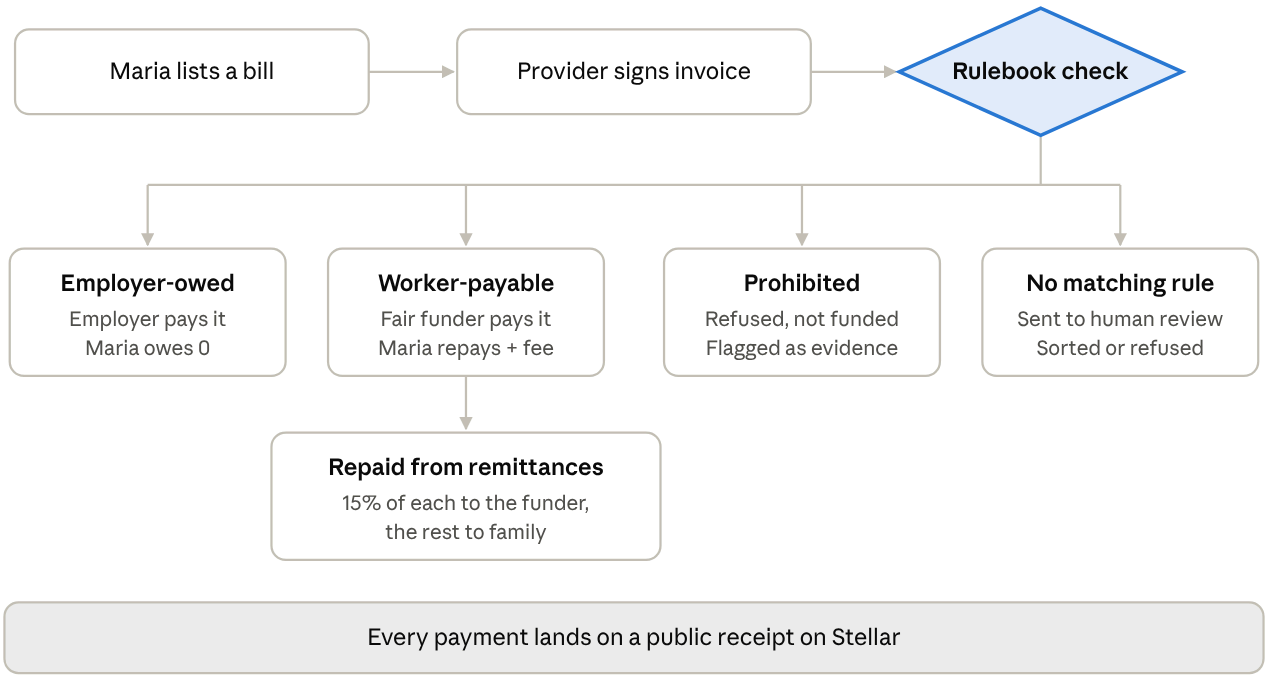
- 80.5% of surveyed OFWs went into debt to finance their placement; two in five felt pressured to use a specific clinic, training center or lender (PCIJ, Migrasia 2022 survey of 961 workers).
- Recruiters steer workers to partner lenders. Complaint documents describe annual rates of 61% to 578% (ICIJ).
- An acceptable training fee is about ₱11,000, but many workers are charged ₱80,000 to ₱90,000 (PCIJ).
- A 2012 union survey of Filipino domestic workers in Hong Kong found an average agency fee of HK$15,378 (about four months' salary); 68% borrowed to pay it (LegCo paper). This is old, advocacy-group data.
- Unpaid debts lead to harassment of families and employers, and some workers have been jailed over post-dated checks signed for these loans (PCIJ).

Root cause: nobody checks who owes each cost before money moves. Payments go through middlemen, mostly in cash, with no record. When employers repay fees afterward, the money arrives late, is based on estimates, and is hard to verify (Transparentem).

Caveats on the evidence: the most extreme figures come from complaints and NGO surveys, not random samples of all workers. Independent confirmation comes from ICIJ, PCIJ and Rappler reporting, and ILO and World Bank research on recruitment costs.

The solution

Laya checks each pre-departure bill against a published rulebook before money moves, sends it to whoever legally owes it, pays the provider directly, and finances only what the worker legitimately owes. Illegal fees are refused. Every payment gets a public receipt on Stellar.



bill flow · rulebook check, four outcomes, repayment

The rulebook check decides who pays; only worker-payable bills become a loan, repaid from remittances.

Roles

Role	Demo name (fictional)	What they do
Worker	Maria	Lists her bills, consents, later sends money home
Provider	Training center, clinic, agency	Bills a cost; must be on Laya's verified list; signs the exact invoice amount
Employer	Hong Kong household	Pays the costs it legally owes (Grant mode)
Mission funder	Fair lender, e.g. a licensed microfinance group	Lends the costs the worker legitimately owes (Loan mode)
Family	Lorna, Maria's mother	Receives remittances; never signs

Role	Demo name (fictional)	What they do
Reviewer / admin	Legal partner in production; our admin in the demo	Maintains the rulebook and provider list; reviews unclear bills

The rulebook and the three piles

A rulebook for one route (Philippines to Hong Kong, domestic work) assigns every bill to a pile. Each rule links to its official source. Laya uses no AI to judge legality.

Pile	Demo example (illustrative)	What Laya does
Employer-owed	Medical exam, 65 USDC	Employer pays the clinic directly; Maria owes 0
Worker-payable	Required training at a fair price, 190 USDC (₱11,000)	Mission funder pays the training center; Maria repays with a flat fee
Prohibited	Placement fee, 1,380 USDC (₱80,000)	Refused and flagged; nobody can fund it
No matching rule	Unknown bill type, price above the fair-price cap, unverified provider	Marked "needs review"; a human reviewer sorts or refuses it

Without Laya, Maria would borrow all 1,635 USDC from a predatory lender. With Laya, the employer pays 65, she borrows 190 on fair terms, and the 1,380 is refused.

Step by step

1. **Maria lists her bills.** Each names a verified provider and an amount. The rulebook sorts it. Maria signs to consent, so nobody can enrol her without her approval.
2. **The provider signs the invoice.** It confirms the exact amount on Stellar. No signature means no payment.
3. **The right party pays the provider directly, in full. A loan is created only when the provider receives payment; if funding fails, the entire transaction reverts and no loan is created.**
    - Employer-owed: the employer pays 65 to the clinic.
    - Worker-payable: the funder pays 190 to the training center. Maria owes 190 plus a flat fee of at most 8% (3% in the demo), so 195.70. The fee is set once and never grows. The 8% cap is our policy, not a legal claim.

    - Prohibited: blocked. Maria gets a record she can show the Philippine labor office.
4. **Maria repays from remittances she sends through Laya.** One transaction splits each remittance. Sending 400 at a 15% share gives 60 to the funder and 340 to Lorna; her balance drops from 195.70 to 135.70, so she finishes in about four months. No remittance means repayment pauses with nothing added. After more than 3 missed periods the loan is flagged overdue and the funder follows up off-chain.
5. **Anyone can check the receipt.** A public page shows each bill, its pile, who paid, how much, the provider's signature, and a fingerprint (SHA-256 hash) of the invoice. A fingerprint proves the invoice is unchanged, not that it is genuine.

Review and disputes

- Bills with no matching rule go to the reviewer's queue automatically; Maria only sees "under review" and can add documents.
- A refusal only means Laya will not finance or route a bill. Nobody is stopped from paying a legal fee elsewhere.
- Providers can appeal with an official fee schedule; the reviewer updates the rule for everyone.
- Real disputes go to Hong Kong's Employment Agencies Administration or the Philippine labor office, which accept complaints from the worker or her representative. Laya supplies the receipt as evidence; it never acts as the court.

Why each party takes part

No step needs anyone to act against their own interest; the open questions are whether they will adopt it.

Party	What they get	Still unproven
Provider	Paid in full, up front, no collection risk	Whether fair-priced providers join the verified list
Employer	A way to pay costs it already owes before departure, with proof, avoiding reimbursement disputes	Whether employers adopt it; the main risk to validate
Mission funder	Small loans (about 190 USDC) for verified costs, repaid in months	Whether enough funders accept a capped flat fee

Party	What they get	Still unproven
Worker	Costs paid without a predatory loan; owes only what is legitimately hers; repayment follows income	Whether she routes remittances through Laya

Employer-funded fee remediation already exists at scale: Impactt reports facilitating over US$111 million in repayments to 86,000+ workers across 210 companies (Impactt). Laya moves that payment before departure and adds proof.

Why Stellar

Stellar settles payments between parties in two countries who share no bank, and gives everyone the same tamper-proof record.

- One transaction splits each remittance between funder and family, for a fraction of a cent.
- Employers, funders, NGOs and auditors can check the public receipt without trusting Laya.
- A conventional database plus a licensed payment provider could do parts of this; the shared, independently verifiable record is the part a database cannot give outsiders. We claim only what the demo shows.

What we build for the hackathon

Most of v1.1 carries over; the contract change is one formula, and the rules live in the app.

- **Contract (Soroban):** keep every function and field name to avoid interface churn (`creditor` now means provider). Change one rule: `legal_amount = principal`, with no 8%-a-year interest. Loan mode keeps the flat fee (at most 8%); Grant mode means employer pays. Remittance split, pause clock and overdue flag are unchanged.
- **Rulebook:** a small table in the app's data for one route, each rule with a source link. The app sorts bills and refuses prohibited ones before anything goes on-chain. In this version the app, not the contract, enforces the rules; we say so.
- **Reviewer:** the existing admin page lists "needs review" bills. Optional if time is short.
- **Schedule:** the current contract can deploy for the Nov 6 integration checkpoint; the formula change ships as a quick redeploy (about 5 minutes with the existing scripts). The client code does not change; golden numbers in the mock and tests do.

- **Demo story:** Maria lists three bills; the employer pays the medical exam (Grant), a funder pays the training (Loan), the placement fee is refused on screen; she remits 400 and the split shows 60 / 340; the public receipt shows every payment.

Open question: an organizer should confirm that an employer-pays flow with a fair-loan step counts for target area 5. Keeping Loan mode prominent in the demo protects the fit.

Known weaknesses and our answers

The remaining weaknesses are adoption and validation risks, not logic holes; challenge fit is the one to settle before building.

#	Weakness	Our answer
1	**Challenge fit:** most of the product is employer-pays and the loan is small	We finance what workers legitimately owe, with repayment that flexes with remittances, and prevent the predatory loans that fill the gap today. Confirm with an organizer
2	**Reach:** a recruiter profiting from illegal fees won't send workers to Laya	We go through the parties who choose the agency: employers and ethical agencies
3	**Employer adoption is unproven**	They already owe these costs; Laya pays before departure with proof. Validation is the next step
4	**Is there a real financing need if the legitimate amount is small?**	₱11,000 is still money a family lacks before the first paycheck; the damage is the ₱80,000 she is pushed to borrow, which we refuse
5	**The rulebook could be wrong** (legal bill refused, or illegal bill passed)	Every rule cites its source; unclear bills go to human review; a refusal never stops anyone paying a legal fee elsewhere; providers can appeal; receipts are evidence for complaints

#	Weakness	Our answer
6	**Who reviews?**	Demo: our admin. Production: legal aid groups that already answer "is this fee legal?" for workers, such as HELP for Domestic Workers and Justice Without Borders (Winrock Remedy Route). Not yet partnered
7	**Funders are a proposed market**	Employers first (grants), mission lenders second. Philippine senators have proposed similar remittance-repaid OFW loans at no more than 6% (Senate); these are proposals, not live programs
8	**Remittance leakage:** repayment only covers remittances sent through Laya	Loans are small and short; production would sit inside a licensed remittance partner
9	**Regulation:** lending and payment licences	Software on licensed partners; testnet only today
10	**Off-platform pressure:** a recruiter can still coerce Maria outside Laya	Laya makes the fair path easy and the unfair one visible; it cannot stop abuse it never sees

Scope and expansion

We start with one route because the rules on who pays are clearest there; only the rulebook changes per route.

- **Starting route:** Filipino domestic workers going to Hong Kong. Hong Kong has an estimated 380,000 foreign domestic workers and about 1,400 employment agencies (SCMP, older figure).
- **Next:** Filipinos to other destinations (Singapore, Taiwan, the Gulf), reusing the Philippine-side rules.
- **Then:** other sending countries. Worker-paid recruitment costs vary widely by route, e.g. about $1,507 India to Saudi Arabia versus $5,637 Pakistan to Saudi Arabia in KNOMAD-ILO surveys (World Bank).

One-line pitch

> Laya makes sure each cost of getting a job abroad is paid by whoever legally owes it: the employer pays what it owes, a fair funder covers what the worker owes and she repays it from money she sends home, and illegal fees are refused, all with a public receipt.

Instructions for reviewers

Please review this brief for a two-week student hackathon, and sort every issue you raise into one of three groups:

1. **Logic holes:** a step that only works if some party acts against its own interest, or a mechanism that cannot do what we claim.
2. **Factual or legal errors:** a claim above that is wrong or overstated. Name the claim and your source.
3. **Validation risks:** things that are plausible but unproven and need evidence (adoption, funders, partners).

Please do not propose an entirely new idea unless you find a logic hole that cannot be fixed within this design. For each issue, say how serious it is for winning the competition and the smallest fix.

Sources

- HK Labour Department: Practical guide for employment of foreign domestic helpers
- HK Government: medical examination fees under the standard employment contract
- HK Government: employer reimbursement of FDH costs
- HK Employment Agencies Administration: what agencies may charge helpers
- SCMP: zero placement fee policy; medical and training fees
- SCMP: Fair Employment Agency; market size
- PCIJ: Migrasia survey and OFW debt
- PCIJ: harassment and jail over loans
- ICIJ: Philippine lenders and labor agents fleece workers
- HK Legislative Council: 2012 union position paper on agency fee overcharges
- Transparentem: partial fee reimbursement in Malaysia

- Impactt: repayment results
- Winrock Remedy Route: overcharging by employment agency
- Philippine Senate: proposed OFW credit assistance loan
- World Bank: recruitment costs presentation (KNOMAD-ILO)